

Tuning Anisotropic Magnon Transport in Multiferroic Oxides by Crystal Symmetry

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Magnon spin current can transport in magnetically ordered insulators without substantial heat dissipation, holding great potential for future spintronic devices. The electric-field control of magnons has garnered recent interest in magnon transport in BiFeO₃ (BFO), a benchmark multiferroic with coupled ferroelectric and antiferromagnetic orders. However, the mechanisms that govern the magnon transport across BFO in heterostructures remain poorly understood. Here, the tuning of anisotropic magnon transport is reported by crystal symmetry in BFO-based system. The single-domain rhombohedral ferroelectric BFO exhibits a spin-cycloid structure and a relatively weak anisotropy of magnon transport. In contrast, a pronounced uniaxial anisotropy of magnon transport is established in the orthorhombic antiferroelectric La-doped BFO with a collinear antiferromagnetic order, which only transmits magnon current with spin polarization along the Néel vector. Moreover, increasing La concentration reduces the propagation length of magnon. The findings pave a new pathway to realize the electric-field-controlled magnon transport for highly efficient magneto-electric spin-orbit (MESO) devices.

spintronics devices.^[1–5] Magnons can deliver spin information in ferrimagnetic or antiferromagnetic insulators without involving moving electrons, as shown in Y₃Fe₅O₁₂,^[6] Fe₂O₃,^[7] Cr₂O₃,^[8,9] and NiO,^[10,11] conferring benefits such as low power consumption, ultrafast response and stability against magnetic fields. Recently, the magnon transport in multiferroic BiFeO₃ (BFO) has attracted increasing interest due to the intrinsic magnetoelectric coupling between ferroelectric and antiferromagnetic (AFM) orders,^[12–20] presenting ideal candidates for the emerging field of electric-field-controlled magnon. Using non-local measurement, the lateral magnon transport in rhombohedral BFO thin films has been extensively studied, which show that the efficiency and anisotropy of lateral magnon transport are correlated to the wave vector *k* of spin-cycloid structure and strain-engineered

domain configuration.^[16,17,19,20] Meanwhile, the vertical magnon transport across BFO in magnetic multilayers has also been investigated, which could enable key functionalities such as magnon torque for deterministic magnetic switching^[10,11,15,16]

1. Introduction

Spin waves excited by collective spin precessions, quantized as magnons, provide promising information carriers for future

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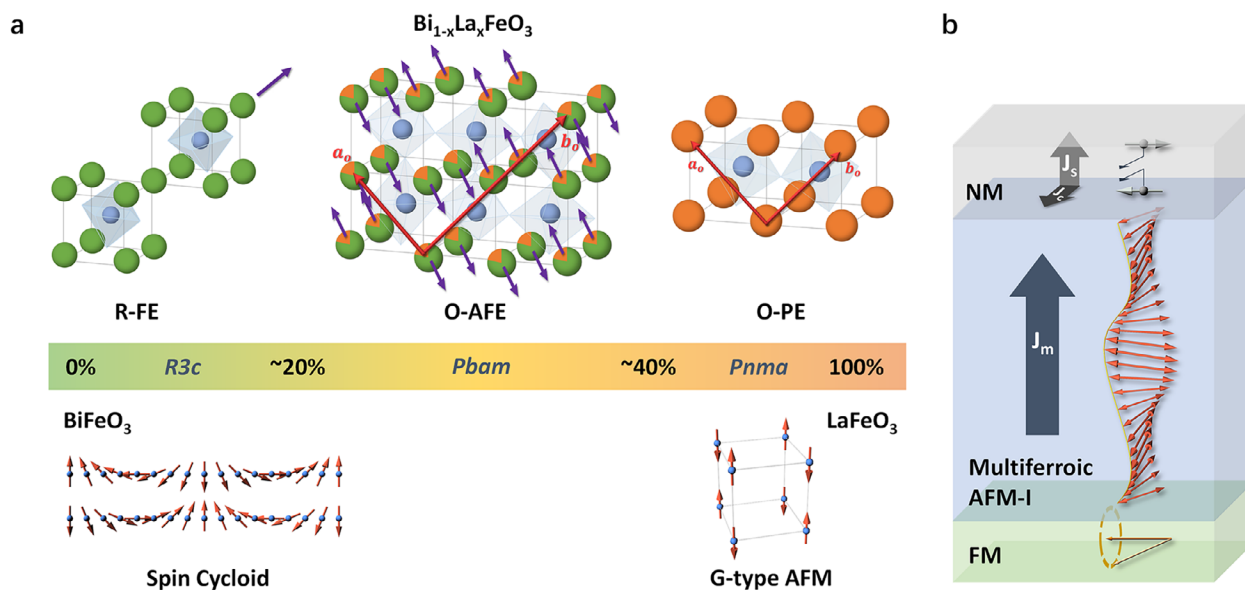


Figure 1. Schematic diagram of vertical magnon transport in the La-doped BiFeO_3 system. a) The phase diagram of crystal symmetry and AFM order in La-doped BFO. In $\text{La}_x\text{Bi}_{1-x}\text{FeO}_3$ system, the crystal structure transforms from rhombohedral ferroelectric phase $R3c$ (0% ~ 20%: R-FE, the purple arrow represents spontaneous polarization P), to the orthorhombic antiferroelectric phase $Pbam$ (20% ~ 40%: O-AFE, red arrows represent crystal axis in the orthorhombic unit cell), and then to the orthorhombic paraelectric phase $Pnma$ (40% ~ 100%: O-PE). The AFM structure transforms from spin-cycloid in the BFO to collinear G-type AFM in LFO. The green, blue and orange atoms represent Bi, Fe and La atoms respectively. b) The schematic of excitation, transmission and detection of vertical magnon current (J_m). The precession of magnetization (yellow arrows) in ferromagnetic (FM) layer leads to spin accumulation at the interface, and excites magnon current (J_m) in multiferroic antiferromagnetic insulator (AFM-I) via spin precession (red arrow). J_m is injected into the normal metal (NM) layer (grey arrows) and converted into current (J_c) by inverse spin Hall effect (ISHE).

and magnon valve^[21,22] for energy-efficient readout. While the non-volatile control of vertical magnon transmission by switching ferroelectric polarization of BFO has been shown,^[15,16] the electric-field-controlled modulation ratio is relatively low (less than 20%), and the underlying mechanism remains elusive. Moreover, besides tuning the ferroelectric polarization, other strategies which can efficiently modulate magnon transport across BFO are highly demanded.

While previous research efforts have mostly focused on the rhombohedral BFO, BFO-based systems exhibit a rich variety of crystal symmetries. A-site substitution in BFO serves as an effective strategy to tune the crystal structure at room temperature, enabling the manipulation of both ferroelectric and magnetic orders.^[23–25] For instance, as shown in **Figure 1a**, the A-site substitution of BFO by rare-earth (such as La) cations leads to the transition from the rhombohedral ferroelectric phase ($R3c$), to the orthorhombic antiferroelectric phase ($Pbam$), and ultimately to the orthorhombic paraelectric phase ($Pnma$).^[26–28] Notably, the AFM structures at the two ends of the phase diagram are markedly different, transitioning from a spin-cycloid structure in rhombohedral BFO^[29–31] to a collinear order in orthorhombic LaFeO_3 (O-LFO).^[32,33] Therefore, the change of crystal symmetry, such as the reversible transition between rhombohedral ferroelectric and orthorhombic antiferroelectric phases,^[34–36] provides another promising route to efficiently tune magnon transport in BFO-based system. However, so far, the AFM structure in the orthorhombic antiferroelectric film has remained insufficiently understood. More importantly, the correlation between the crystal symmetry and magnon transport in BFO-based systems has not yet been explored.

In this work, we demonstrate crystal-symmetry-dependent anisotropy in magnon transport in multiferroic La-doped BFO system. Rhombohedral ferroelectric BFO (R-BFO) and orthorhombic antiferroelectric La-doped BFO (O-LBFO) with structural mono-domain are synthesized to avoid complex contributions from multiple domains. A spin-cycloid structure is identified in R-BFO, while O-LBFO exhibits a collinear AFM monodomain state. Angular dependent spin pumping measurements, performed in BFO-based heterostructures (see **Figure 1b**), reveal distinct anisotropy in vertical magnon transport between R-BFO and O-LBFO. R-BFO shows a relatively weak anisotropy in magnon transport with respect to the spin polarization of the injected spin current. In contrast, O-LBFO exhibits a strong uniaxial anisotropy, only allowing magnon current transmission when the spin polarization is parallel to the Néel vector. Furthermore, we find that the increase of La concentration reduces the magnon propagation length in $\text{La}_x\text{Bi}_{1-x}\text{FeO}_3$, likely correlated to the modulation of exchange interaction strength. Our results provide new insights into the mechanisms that control magnon transport in BFO-based system, and offers design principles for developing electric-field-controlled magnon devices.

2. Results

2.1. Crystalline Monodomain in R-BFO and O-LBFO

As illustrated in **Figure 1b**, the vertical magnon current J_m is excited by the precession of ferromagnetic (FM) layer, transmits through the multiferroic AFM insulator (AFM-I) layer, and is subsequently detected via the inverse spin Hall effect (ISHE) in

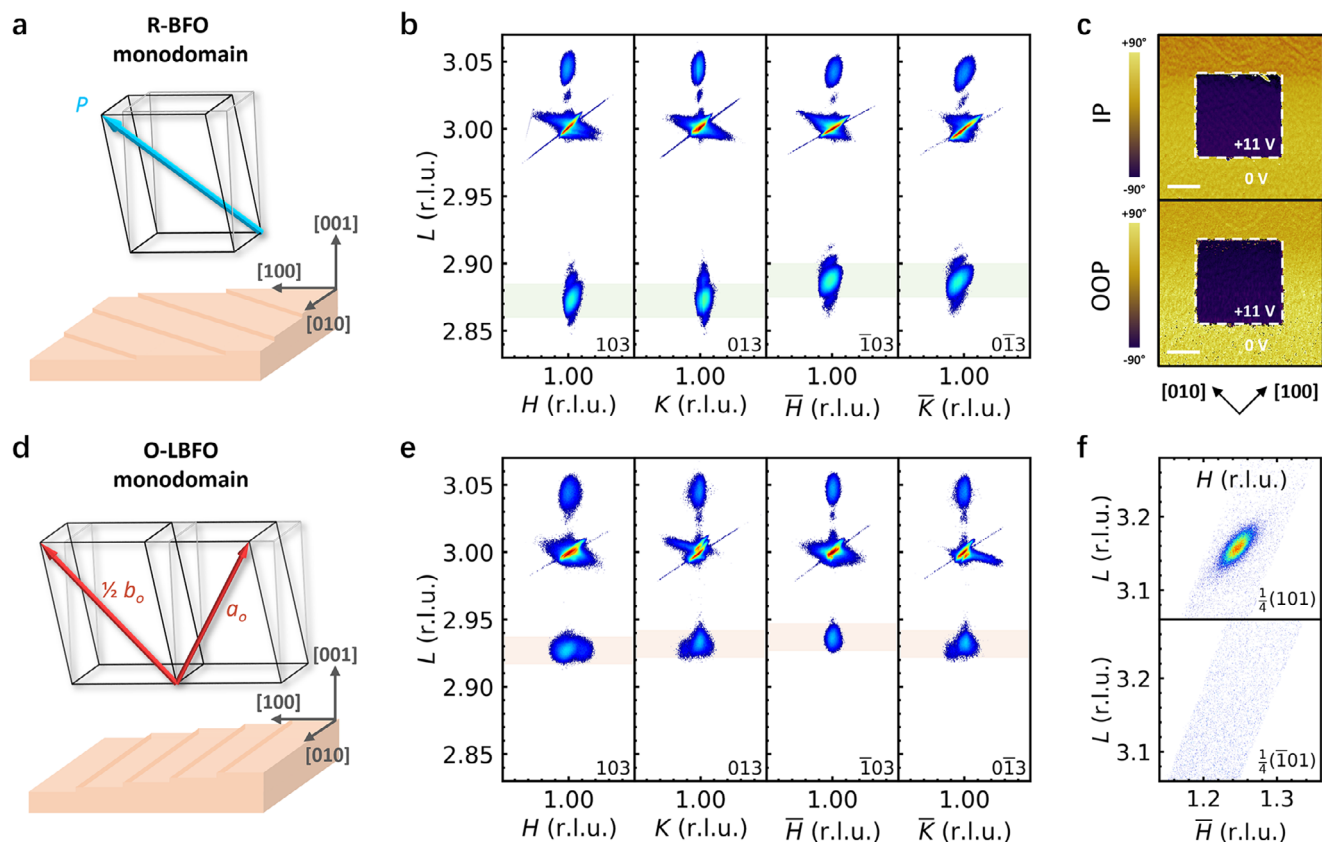


Figure 2. Realization of monodomain state in R-BFO and O-LBFO. a) Schematic diagram of R-BFO monodomain induced by [110] miscut substrate. Only the domain with ferroelectric polarization $\mathbf{P} \parallel [111]$ (blue arrow) is retained. b) Reciprocal space mapping (RSM) around (103) Bragg peaks of LSMO (22 nm)/R-BFO (30 nm) on STO. The R-BFO (103) peaks are highlighted with a green background. c) In-plane and out-of-plane piezo-force microscopic (PFM) phase image of $\mathbf{P}$ switching by applying +11 V voltage (inside the white boxes), with the cantilever along [110]. Similarly, d) and e) show the schematic diagram of O-LBFO monodomain induced by [100] miscut substrate and the corresponding RSM of four (103) Bragg peaks of LSMO (22 nm)/O-LBFO (30 nm) sample. The O-LBFO (103) peaks are highlighted with an orange background. f) The RSM of antiferroelectric superlattice $1/4(101)$ and $1/4(\bar{1}01)$ peaks.

the normal metal (NM) layer.^[37,38] Here, we use $\text{La}_{0.67}\text{Sr}_{0.33}\text{MnO}_3$ (LSMO) with in-plane magnetic anisotropy and a relative high Curie temperature ($T_c \sim 340$ K) as the FM layer, and Pt as the NM layer. Three representative oxides, i.e., R-BFO, O-LBFO with nominal 28% La concentration, and O-LFO, were chosen to reveal the vertical magnon transport. Bulk R-BFO shows a spontaneous ferroelectric polarization $\mathbf{P} \sim 100 \mu\text{C}/\text{cm}^2$ along $[111]_{\text{pc}}$ (pc: pseudo-cubic, used in this work unless otherwise stated).^[39] The Dzyaloshinskii-Moriya interaction (DMI) arising from spin-orbit coupling and broken spatial inversion symmetry leads to the formation of spin-cycloid structure in R-BFO with propagation vector $\mathbf{k} \perp \mathbf{P}$.^[30,40,41] In contrast, bulk O-LBFO exhibits an antiferroelectric order within the $\mathbf{a}_o\mathbf{b}_o$ plane (o: orthorhombic) and likely the G-type AFM order.^[27,28]

High-quality heterostructures comprised of 22-nm-thick LSMO, R-BFO/O-LBFO/O-LFO with varied thicknesses, and 5-nm-thick Pt are deposited on SrTiO_3 (STO) substrates (see Experimental Section for growth). The θ -2 θ X-ray diffraction (XRD) scans are shown in Figure S1 (Supporting Information). The LSMO/Pt bilayers are also synthesized as references. To facilitate the growth of crystalline monodomain films for R-BFO,

O-LBFO and O-LFO, (001)-oriented vicinal STO substrates are used.

As shown in Figure 2a, the vicinal STO substrate with a 4° mis-cut angle along in-plane [110] direction breaks the four-fold rotation symmetry of the polarization variants, leading to the monodomain R-BFO films.^[42] This is verified by measuring the reciprocal space mapping (RSM) of four (103) peaks of the LSMO/R-BFO heterostructures on the vicinal substrates. As shown in Figure 2b, the (103) and (013) peaks of R-BFO shift downwards while the $(\bar{1}03)$ and $(0\bar{1}3)$ peaks shift upwards along the L-direction, supporting the formation of monodomain R-BFO. The ferroelectric monodomain state is further confirmed by using piezo-force microscopic (PFM) imaging (see Figure 2c). The as-grown R-BFO exhibits a uniform contrast in both in-plane and out-of-plane phase images and predominantly undergoes 180° polarization switching under applied positive voltage.

For O-LBFO and O-LFO, vicinal STO substrates with a 4° mis-cut angle along in-plane [100] direction are used. Similar to the previous report on O-LFO film, the vicinal direction along [100] induces a tilt which favors only one structural domain with the $\mathbf{a}_o\mathbf{b}_o$ (o: orthorhombic) plane perpendicular to the film plane and

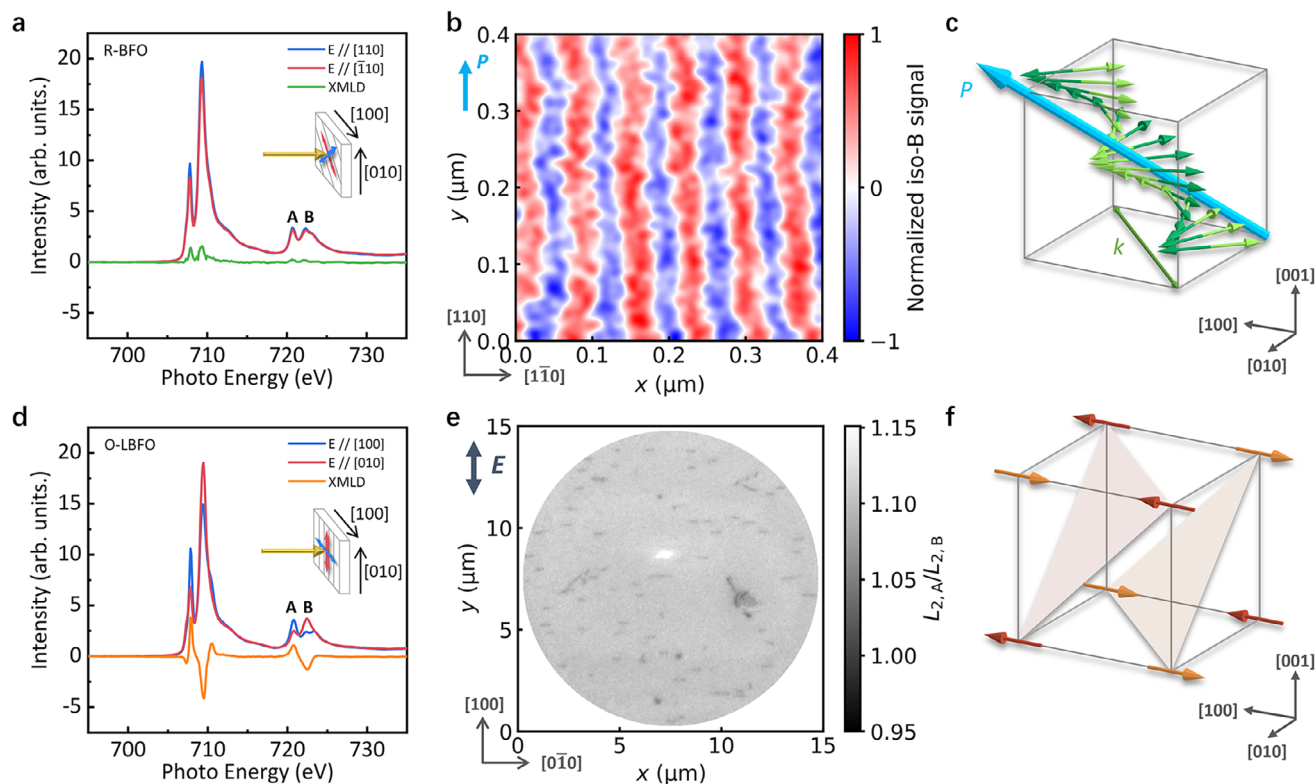


Figure 3. Magnetic structure of monodomain R-BFO and O-LBFO. a) The XMLD spectra of LSMO (22 nm)/R-BFO (10 nm) sample with normal incidence at $E // [110]$ (blue) and $E // [\bar{1}10]$ (red) using the total-electron-yield (TEY) mode. The inset shows the test configuration. b) The NV image of LSMO (2 nm)/R-BFO (10 nm) sample. We use the LSMO (2 nm) with a largely reduced thickness to avoid the influence of stray fields from LSMO. The blue arrow represents the direction of ferroelectric polarization P . c) The schematic of spin-cycloid structure with $k \perp P$ in R-BFO. d) The XMLD spectra of LSMO (22 nm)/O-LBFO (10 nm) sample with normal incidence at $E // [100]$ (blue) and $E // [010]$ (red). e) The XMLD-PEEM image at $E // [100]$ of LSMO (22 nm)/O-LBFO (10 nm) sample. The dark arrow represents the direction of electric-field vector E of incident X-ray. f) The corresponding G-type AFM structure in O-LBFO with $L \perp$ step edge.

the b_o -axis along [010] (see Figure 2d).^[33] Figure 2e shows the RSM of four (103) peaks for the LSMO/O-LBFO heterostructures. Along the vicinal direction [100], the (103) and ($\bar{1}03$) peaks exhibit opposite shift in L-directions, while no shift is observed for the (013) and ($0\bar{1}3$) peaks, confirming the formation of monodomain O-LBFO. Note that small satellite peaks are observed near the main Bragg peaks of O-LBFO, which are likely due to strain relaxation. We further measure the antiferroelectric superlattice peak $1/4(101)$ in the O-LBFO films, which originates from the antiparallel displacement of A-site cations in $PbZrO_3$ -like P_{6mm} structure. As shown in Figure 2f, the superlattice peak $1/4(101)$ is clearly observed in our film, while the $1/4(\bar{1}01)$ peak is below the measurement threshold. These results further confirm the formation of monodomain O-LBFO with the antiferroelectric order.

2.2. Magnetic Structure of Monodomain R-BFO and O-LBFO

Having confirmed the structural monodomain of both R-BFO and O-LBFO, we proceed to examine their magnetic structures. Angle-dependent soft X-ray absorption spectroscopy (XAS) and X-ray magnetic linear dichroism (XMLD) measurements are carried out at Fe-L edges at room temperature with normal incidence, by varying the X-ray polarization vector E with respect to

the crystalline direction. Figure 3a shows the XAS spectra of the R-BFO with vector E parallel to ($E // [\bar{1}10]$) and perpendicular to ($E // [110]$) the steps of vicinal substrates, exhibiting a low XMLD signal as defined by the difference of the XAS spectra. Previous studies have shown that the relative intensity of peaks A and B at the Fe- L_2 edge, as marked in Figure 3a, reveals the orientation of Néel vector L .^[43–45] To fully resolve the magnetic structure, we measure angle-dependent XAS with E rotating within the (001) plane,^[45] where the angle between E and [100] is denoted as θ_E (see Figure S2, Supporting Information). As θ_E increases from 0° to 180° , the peak intensity ratio of $L_{2,A}/L_{2,B}$ remains almost the same in R-BFO. The absence of XMLD signal in R-BFO could be ascribed to the presence of spin-cycloid structure, which is further verified by using nitrogen-vacancy (NV) diamond-based scanning probe magnetometry.^[30] As shown in Figure 3b, the NV images suggest that the sample exhibits a uniform spin cycloid structure with a period of ≈ 70 nm and the propagation vector k perpendicular to the polarization P (see fast Fourier transform (FFT) analysis in Note S3, Supporting Information), further confirmed by comparing the topography, PFM, and NV images of the same R-BFO sample (see Figure S3, Supporting Information).^[46] These findings are consistent with previous neutron diffraction and XMLD results, which reveal that the R-BFO on vicinal STO substrate exhibits the spin-cycloid structure with its cycloid plane

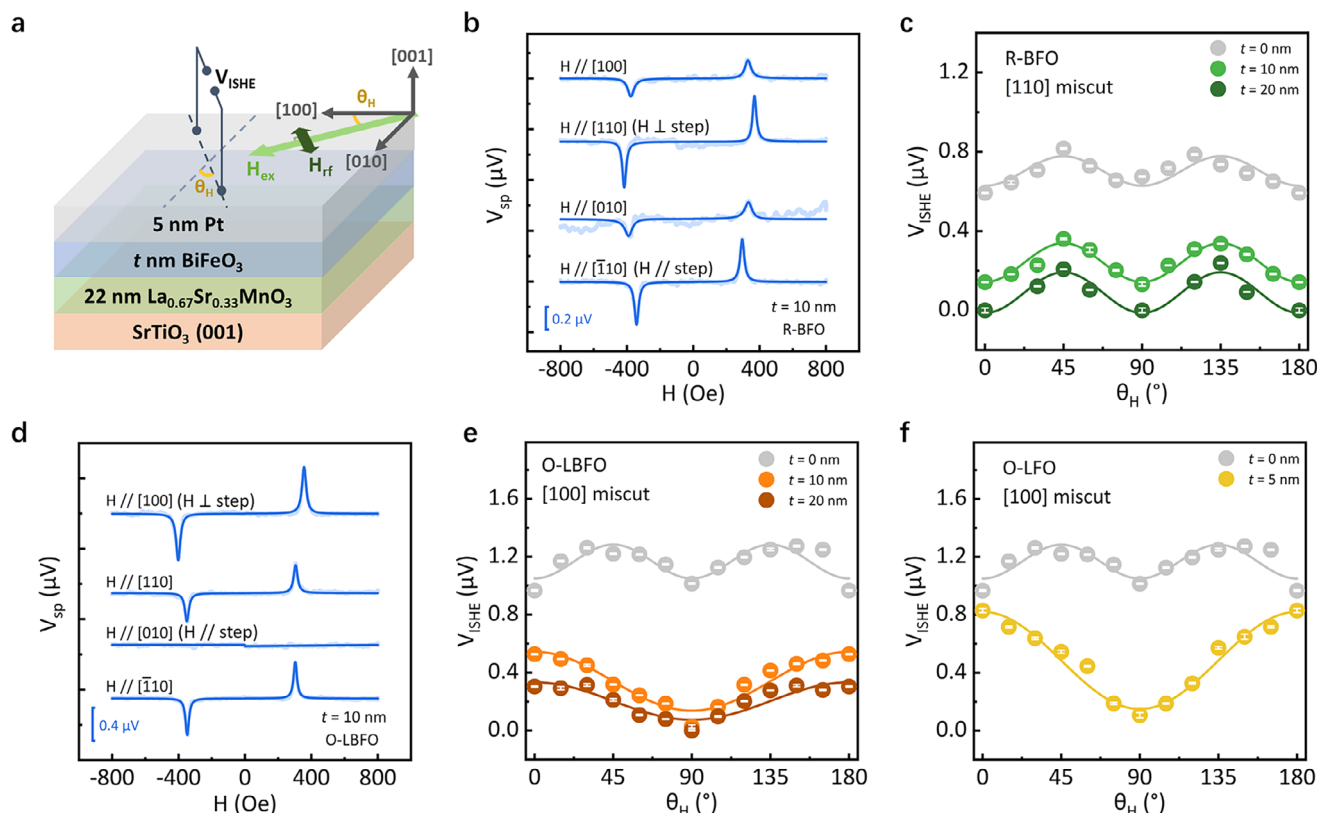


Figure 4. Angle-dependent spin pumping measurement in monodomain R-BFO and O-LBFO. a) Experiment setup of angle-dependent spin pumping measurement to detect V_{ISHE} induced by vertical magnon current. b) $V_{\text{sp}} - H_{\text{ex}}$ (H) spectra of tri-layers with $t = 10$ nm monodomain R-BFO, with $H // [100]$, $[110]$, $[010]$ and $[\bar{1}10]$. The blue line is Lorentzian fitting of original data (light blue). c) The angle-dependent V_{ISHE} with θ_H varying from 0° to 180° (solid dots) of heterostructures with $t = 0$, 10 and 20 nm monodomain R-BFO, which follow the $\cos^2(2\theta_H)$ fitting (solid line). d) $V_{\text{sp}} - H_{\text{ex}}$ (H) spectra of tri-layers with $t = 10$ nm monodomain LBFO with $H // [100]$, $[110]$, $[010]$ and $[\bar{1}10]$. e) The angle-dependent V_{ISHE} with θ_H varying from 0° to 180° (solid dots) of heterostructures with $t = 0$, 10 and 20 nm O-LBFO and fitting (solid line). f) The angle-dependent V_{ISHE} of heterostructures with $t = 0$ and 5 nm monodomain O-LFO and fitting. The error bars represent standard deviations arising from uncertainties in the Lorentzian fitting.

nearly parallel to the film surface.^[31] Figure 3c schematically shows the magnetic structure of R-BFO films, where the Fe spins form cycloid structure with $\mathbf{k}$ parallel to the substrate steps.

By contrast, as shown in Figure 3d, the O-LBFO exhibits a significant XMLD signal between $\mathbf{E}$ parallel to ($\mathbf{E} // [010]$) and perpendicular to ($\mathbf{E} // [100]$) to steps. Detailed angle-dependent XAS measurements of O-LBFO further reveal a $\cos^2(\theta_E)$ angular dependence of the $L_{2,A}/L_{2,B}$ intensity ratio (see Figure S2, Supporting Information). These results imply the presence of a collinear AFM order with in-plane projection of Néel vector $\mathbf{L}$ along $[100]$ ($\mathbf{L} \perp \text{step}$) in the monodomain O-LBFO, similar to the O-LFO on vicinal STO as previously reported.^[33] Note that our results are different from the previous report for O-LBFO on STO, in which the absence of XMLD is likely due to the formation of multiple domains.^[28] To spatially resolve the AFM domain structure, we conduct X-ray magnetic linear dichroism-photoemission electron microscopy (XMLD-PEEM) measurements on the O-LBFO sample. Figure 3e shows the mapping of $L_{2,A}/L_{2,B}$ ratio taken with incident X-ray along $[010]$ and $\mathbf{E} // [100]$. The XMLD-PEEM image shows a nearly uniform contrast for the majority of the sample ($\sim 85\%$), revealing a nearly AFM monodomain state with $\mathbf{L} // [100]$. This is verified by varying the incident X-ray direction

(see further discussions on the XMLD-PEEM results in Figure S4, Supporting Information). Figure 3f schematically shows the magnetic structure of O-LBFO film, exhibiting a collinear AFM order with in-plane projection of $\mathbf{L}$ perpendicular to the steps.

2.3. The Correlation Between Magnon Transport and Crystal Symmetry

Next, we investigate the vertical magnon transport across monodomain R-BFO, O-LBFO, and O-LFO using spin pumping measurements. Figure 4a shows the sample structure and experimental configuration. An external magnetic field H_{ex} is applied within the film plane to saturate the magnetization of LSMO. A small oscillating magnetic field H_{rf} , generated by a microwave source, excites the ferromagnetic resonance of LSMO, which injects a pure spin current with spin polarization σ along the LSMO magnetization into the BFO layer. The magnon current transmitted through the BFO layer is then detected by the inverse spin Hall effect (ISHE) in the Pt, which converts a spin current into an electron current. The amplitude of the ISHE voltage (V_{ISHE}), obtained

by the Lorentzian fitting of each $V_{sp} - H$ spectrum (see more details in Note S5, Supporting Information), reflects the magnitude of magnon current transmitted through BFO.^[37,38] By changing the relative angle (denoted by θ_H) between H_{ex} (thereby spin polarization of injected spin current) and crystallographic direction ([100]) through rotating the sample, we obtain the angular dependence of vertical magnon transmittance. We first measure the V_{ISHE} signal with H_{ex} (H) aligned along [100], [110], [010] and $\bar{[110]}$ at 300 K for a reference LSMO/Pt bilayer (no BFO layer, $t = 0$ nm). As shown in Figure S5 (Supporting Information), the ferromagnetic resonance field appears at ~ 400 Oe, and the sign of V_{ISHE} changes as reversing H_{ex} . The magnitude of V_{ISHE} exhibits fourfold rotation symmetry in the reference LSMO/Pt, with larger values along [110] and $\bar{[110]}$ directions, likely due to the magnetic anisotropy of LSMO films.

The LSMO/R-BFO/Pt tri-layer (with a 10 nm R-BFO layer) preserves this fourfold anisotropy, as shown in Figure 4b. A relatively large V_{ISHE} is observed at both $H \parallel [110]$ (perpendicular to step edge, $H \perp$ step) and $H \parallel \bar{[110]}$ (parallel to step edge, $H \parallel$ step), which reduces when the magnetic field is along [100] and [010] directions. Note that the V_{ISHE} vanishes in control samples by replacing R-BFO with non-magnetic STO, which reveals the critical role of magnon transport across BFO and excludes other non-magnon-related origins such as spin rectification or charge leakage (see Figure S6, Supporting Information). This is also supported by the power-dependent V_{ISHE} (see Figure S7, Supporting Information). We further measure the spin pumping signals of the samples by systematically varying θ_H . Figure 4c summarizes the angular dependences of V_{ISHE} for LSMO/Pt bilayer ($t = 0$ nm) and LSMO/R-BFO/Pt tri-layers (with $t = 10$ and 20 nm) on vicinal STO with a mis-cut along [110]. All samples exhibit similar fourfold rotation symmetry, well-described by a $\cos^2(2\theta_H)$ fit (solid lines). The decrease of V_{ISHE} as increasing t is due to the attenuation of magnon transport across BFO. These results indicate that R-BFO shows a relatively weak anisotropy in vertical magnon transport (see further discussions in Figure S8, Supporting Information).

In contrast, O-LBFO induces a distinctly different angular dependence in spin pumping signals. As shown in Figure 4d, a strong V_{ISHE} signal is observed when $H \parallel [100]$ ($H \perp$ step), while the signal almost fully vanishes when $H \parallel [010]$ ($H \parallel$ step). Figure 4e shows the angle-dependent V_{ISHE} of the LSMO/Pt bilayer ($t = 0$ nm) and LSMO/O-LBFO/Pt tri-layers (with $t = 10$ and 20 nm) on vicinal STO with a mis-cut along [100]. While the LSMO/Pt bilayer still shows a fourfold rotation symmetry, the tri-layers with O-LBFO exhibit a distinct twofold rotation symmetry that follows a $\cos^2(\theta_H)$ dependence. To be more specific, the maximal (minimal) V_{ISHE} occurs at $\theta_H = 0^\circ$ (90°) when the injected spin polarization is parallel (perpendicular) to the Néel vector L , consistently observed at different resonance frequencies of LSMO (see Supporting Information Figure S9, Supporting Information). These results demonstrate that the O-LBFO exhibits a pronounced anisotropy of vertical magnon transport, acting as a magnon filter for the injected spin current. Similar behavior is also observed in monodomain O-LFO (see Figure 4f), suggesting the critical role of crystalline symmetry. Note that the distinct anisotropy between bilayer and tri-layer samples further verifies that the V_{ISHE} is determined by magnon transport through the multiferroic layer.

To further explore the influence of La substitution, we measure the thickness dependence of V_{ISHE} for the tri-layers with monodomain R-BFO, O-LBFO and O-LFO (see results in Figures S10 and S11, Supporting Information), with the magnetic field applied along the maximal transmission direction ($H \perp$ step). Figure 5a summarizes the normalized V_{ISHE} for the three sets of samples, which all show a decay of V_{ISHE} by increasing the thickness. In R-BFO-based sample, a clear V_{ISHE} signal above the noise level (~ 100 nV) is observed up to $t = 60$ nm. In contrast, the V_{ISHE} signal almost vanishes for O-LBFO at $t = 40$ nm and O-LFO at $t = 20$ nm. By fitting with the exponential decay model, the diffusion lengths (λ_m) are determined to be 24.9, 15.7 and 9.1 nm for R-BFO, O-LBFO, and O-LFO (see Note S10, Supporting Information). These results suggest a decrease of magnon propagation length by increasing the La concentration in the $La_xBi_{1-x}FeO_3$ system.

2.4. Mechanism of Vertical Magnon Transport in BFO System

The correlation between angle-dependent magnon transmittance and crystal symmetry of BFO-based system is schematically illustrated in Figure 5b,c. For vertical magnon current with spin polarization σ , the R-BFO exhibits a weak anisotropy by varying the directions of σ . In contrast, O-LBFO and O-LFO only allow magnon current transmission when σ is parallel to the in-plane projection of Néel vector, exhibiting a strong uniaxial anisotropy. The phenomenological model for the vertically transmitted magnon current can be expressed as:^[7,16]

$$J_m = (G_L \oint [L(r) \cdot \sigma]^2 dr + G_m \oint [m(r) \cdot \sigma]^2 dr) \cdot J_{injection} \quad (1)$$

where σ is the spin polarization of the injected spin current, G_L and G_m are the coefficients determined by spin conductance that is closely correlated to the exchange interaction/anisotropy energy of the AFM order, $L(r)$ and $m(r)$ are the profile functions of the Néel vector and net magnetization. The net magnetization m is relatively low in the BFO-based systems and thus we mainly consider the first term for the vertical magnon transport.

The distinct anisotropy between R-BFO and O-LBFO mainly arises from the different AFM structures. As shown in Figure 5b, the R-BFO exhibits the spin-cycloid structure with its cycloid plane almost parallel to the film plane. Therefore, the integration of the $[L(r) \cdot \sigma]^2$ term remains a constant coefficient as varying the direction of σ with respect to the crystal axis. The weak anisotropy of R-BFO could be due to the G_L term, correlated to its crystal symmetry. In contrast, the O-LBFO shows the collinear AFM with a well-defined L along [100] direction. As a result, the $[L(r) \cdot \sigma]^2$ term shows a clear $\cos^2(\theta_H)$ angular dependence as varying the direction of σ , leading to the uniaxial anisotropy with maximum value when $L \parallel \sigma$ (see analyses in Note S11, Supporting Information).

Moreover, Figure 5a shows that the magnon diffusion length reduces as increasing La concentration. Given that O-LBFO and O-LFO share the similar orthorhombic structure, this decreasing trend suggests that the reduction of magnon propagation length is likely governed by the La substitution ratio, rather than the crystal symmetry or cation disorder. We believe this could be due to the modulation of spin conductance coefficient

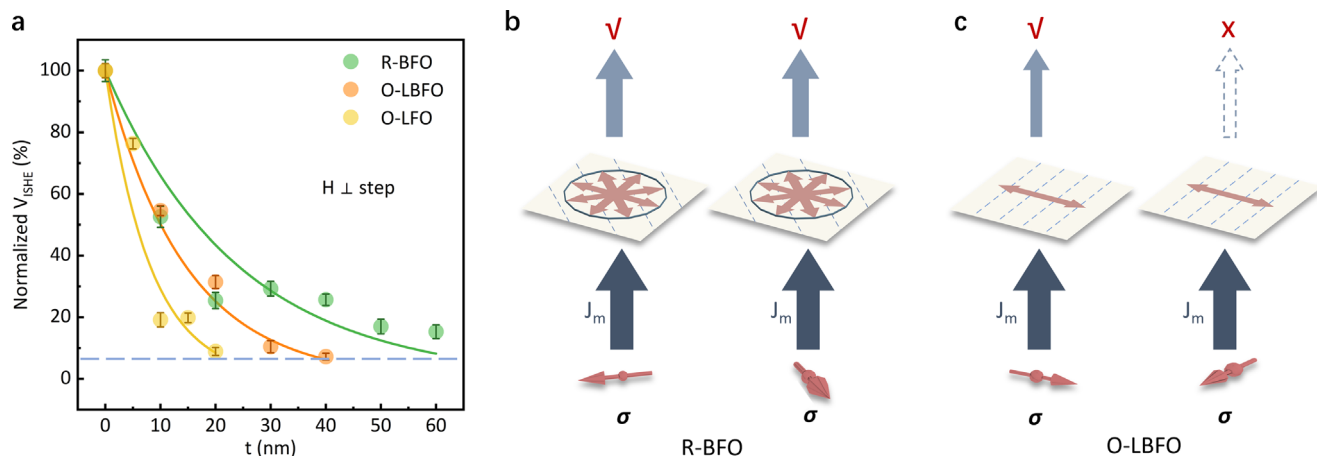


Figure 5. Tuning of vertical magnon transport by crystal symmetry. a) Thickness-dependent normalized V_{ISHE} signals of heterostructures with monodomain R-BFO (green dots), monodomain O-LBFO (orange dots) and monodomain O-LFO (yellow dots). The error bars are standard deviations based on three independent measurements. The solid line represents the fitting using diffusion model with exponential decay. The dash line marks the upper bound of the measurement noise floor. b) Schematic diagram of vertical magnon transport in R-BFO. The in-plane projection of the spin-cycloid forms a circle, allowing J_m with all directions of spin polarization to pass through. Blue arrows represent the relative magnitude of the magnon current J_m . c) Schematic diagram of vertical magnon transport in O-LBFO. The in-plane projection of the spin-cycloid forms a circle, allowing J_m with all directions of spin polarization to pass through. Blue arrows represent the relative magnitude of the magnon current J_m . The in-plane projection of the Néel vector (L) is along the vicinal direction, allowing only magnon current with spin polarization $\sigma \parallel L$ (left panel) to get through, whereas that with $\sigma \perp L$ (right panel) is fully blocked.

G_L , which is closely correlated to the exchange interaction and anisotropy energy strength. Due to different cation size, La substitution could modulate the exchange interaction strength between neighboring Fe cations by changing the bond length and angle, as evidenced by the increase of Néel temperature upon La substitution.^[29,47,48] This stabilizes the antiparallel alignment of neighboring Fe spins, which could impede the collective spin precession that enables the magnon propagation.

3. Conclusion

In summary, we have demonstrated the tuning of magnon transport in BFO-based system by crystal symmetry and have established its correlation with the AFM structures. The R-BFO with spin-cycloid structure shows a relatively weak anisotropy of magnon transport. In contrast, O-LBFO exhibits a strong uniaxial anisotropy, acting as a spin-current filter that allows magnon transmission only when the spin polarization aligns parallel to L . Moreover, the magnon propagation length reduces as increasing La concentration, revealing the role of exchange interaction strength on magnon transport. Our results advance the understanding of magnon transport mechanisms in multiferroic BFO system, and offer design principles for realizing electric-field control of magnon transport.

Importantly, our results may lay the foundation for developing efficient magneto-electric spin-orbit (MESO) devices. Previous studies have demonstrated the reversible control between rhombohedral and orthorhombic phases by applying an electric-field in BFO-based system.^[34–36] Building upon this, our work suggests that the distinct angular and thickness dependencies of vertical magnon transport across these phases could be exploited to significantly enhance the electric-field-controlled switching ratio of magnon transmission (see further discussions in Note S12; Figure S12, Supporting Information).

4. Experimental Section

Sample Growth: Pulsed laser deposition (PLD) was used with a KrF excimer laser (248 nm) to grow high-quality heterojunctions epitaxially on SrTiO₃ (001) substrate. The substrate temperature was 750 °C in an oxygen partial pressure of 200 mTorr for LSMO. The R-BFO, O-LBFO, and O-LFO layer were grown by using the ceramic targets with nominal compositions of BiFeO₃, La_{0.28}Bi_{0.72}FeO₃, and LaFeO₃, at 700 °C and 100 mTorr oxygen partial pressure. The laser energy density was $\approx 1.0 \text{ J cm}^{-2}$ and repeated at 5 Hz. The distance between substrate and target was 60 mm. After the deposition, the sample was annealed at 150 Torr oxygen for 20 min and then cooled to room temperature at a rate of 10 °C min^{−1}. Subsequently, the sample was transferred to a magnetron sputtering chamber with a background vacuum of 1×10^{-8} Torr for the deposition of Pt layer at an Ar pressure of 0.3 Pa.

Structural and Morphology Characterization: The X-ray diffraction (XRD) and RSM measurements were characterized by high-resolution X-ray diffraction (XRD, Malvern PANalytical Empyrean) with Cu-K α radiation ($\lambda = 0.1542 \text{ nm}$). The PFM images were obtained by MFP 3D Infinity atomic force microscopy from Oxford Instrument, Asylum Research.

Spin-Pumping Measurement: Spin pumping was performed in Physical Property Measurement System (PPMS) chamber. A microwave source Keysight E8257D provides a 4 GHz microwave with nominal power of 20 dBm to the coplanar waveguide for generating a ferromagnetic resonance of LSMO during the spin pumping measurements. The microwave signal source was modulated by a 1 Vrms, 1713 Hz AC voltage. The V_{ISHE} signals between the pads were detected by an SR830 lock-in amplifier synchronized to the first harmonic. All data were measured at 300 K.

XAS and XMLD: The angle-dependent XAS and XMLD measurements were carried out at BL07U beamline at Shanghai Synchrotron Radiation Facility (SSRF) in Shanghai, China, with a beam spot size of $\approx 100 \times 50 \text{ }\mu\text{m}$ in the energy range 690–740 eV. The incident angle was normal to the sample surface. A single crystal $\alpha\text{-Fe}_2\text{O}_3$ was measured for energy calibration. The XMLD-PEEM data was obtained at BL17SU beamline at SPring-8 in Japan. Monochromatized X-ray beam at Fe-L₂ edge were directed onto the sample at a grazing angle of 30 degrees. All data were measured at 300 K.

NV Microscopy: BFO samples were measured at room temperature using a commercial scanning NV microscope (QDAFM, CIQTEK). A diamond probe with single NV center was used to increase photon collection efficiency (CIQTEK). A small magnetic field with a projection along the NV

center axis was applied to increase the contrast of the photoluminescence (PL) intensity obtained from electron spin resonance (ESR) spectroscopy. The NV images were collected in the “dual-iso-B” mode.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

X.Q. and H.C. contributed equally to this work. D.Y. and T.N. conceived the idea and supervised the project. X.Q., H.C., and Y.Y. grew the films. X.Q. and H.C. performed the spin transport measurements and data analysis. X.Q. performed the XMLD measurement with assistant of Y.Z. and F.Z. X.Q., Y.Z., J.M., and X.J. performed the XMLD-PEEM measurements, supervised by T.O.X.Y. and X.Q. performed the NV measurements. X.Q., D.Y., and T.N. wrote the manuscript. All the authors discussed the results and contributed to the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

crystal structure, magnon current, multiferroics, spin pumping

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